CM3 : Geometric Algorithmique

Reminder

CM1 : Some important problems
- → the experts problem (learning theory)
- → discrepancy problem (algorithms)

CM2 : Probability background.

CM3 : - finish probability background
- geometry background.

Next week : join with the Berlin lectures.

Random variable: X ← real values

most important information: weighted average value of X → expectation $E[X]$

$$E[X_1 + X_2] = E[X_1] + E[X_2]$$

Super important fact:

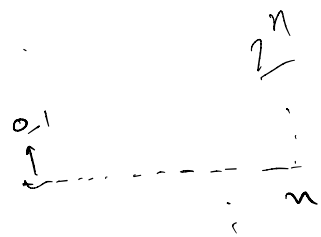
↓

Critical because it allows us to "decompose" a complicated random variable in easier pieces.

Example: toss a coin n times.

X: # of times a tail appears

$E[X]$



linearity of expectation

$$E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i] = \frac{n}{2}$$

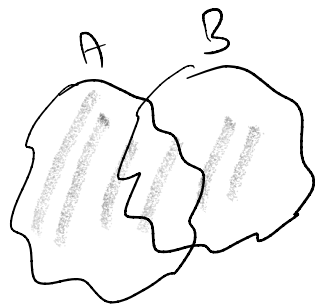
$$\sum_{i=0}^{n} \frac{\binom{n}{i}}{2^n} \cdot i = \frac{n}{2}$$

X_i: 1 if i-th toss is a tail

Second important fact : Union bound

$$\Pr[\mathcal{E}_1 \cup \mathcal{E}_2] \leq \Pr[\mathcal{E}_1] + \Pr[\mathcal{E}_2]$$

in combinatorial view :



$$|A \cup B| \leq |A| + |B|$$

exact calculation. $|A \cup B| = |A| + |B| - |A \cap B|$

complicated with many $\mathcal{E}_1 \dots \mathcal{E}_n$. rarely used.

Application:



$X = \{1,2,3,4,5\}$

$R = \{\{1,3\}, \{2,4\}, \{1,2,3\}, \{2,3,4\}, \{5\}\}$

(X, R) — X: n elements; R: m sets.

$\{x_1, \dots, x_n\}$

$\varepsilon \in (0,1]$

$\forall S \in R: |S| \geq \varepsilon \cdot n$

optimization also hitting set problem.

Question: Find a $N \subseteq X$ of small size such that $\forall S \in R: N \cap S \neq \emptyset$.

Algorithm:

$N = \emptyset$

For $t = 1 \dots n$:

add x_i to N with prob. p

Claim: With prob. $\geq 1 - \frac{1}{n}$, for each $S \in R$, we have that $N \cap S \neq \emptyset$.

"with high prob."

Plan for the analysis:

1. we compute the probability of failure ($N \cap S = \emptyset$) for each $S \in R$.
2. we use the union bound to upper bound the total failure probability.

(1) Fix $S \in R$.

$$\Pr\left[\begin{array}{l}N \text{ does not} \\ \text{contain any element of } S\end{array}\right] = (1-p)^{|S|}$$

$1+x \le e^x$ $\rightarrow$ $\le e^{-p \cdot |S|}$

since $|S| \ge \varepsilon n$ $\rightarrow$ $\le e^{-p\,\varepsilon n}$.

(2) $\Pr\left[\begin{array}{l}N \text{ does not contain} \\ \text{any element from } \underline{\text{some}}\ S \in R\end{array}\right]$ $\overset{\text{union bound}}{\le} \sum_{S \in R} e^{-p\,\varepsilon n}$

$= m \cdot e^{-p\,\varepsilon n}$.

set p s.t. $m \cdot e^{-p\,\varepsilon n} \le \frac{1}{n}$

$$p = \frac{1}{\varepsilon n} \ln(mn)$$

$$m \cdot e^{-\ln(mn)} = \frac{m}{mn} = \frac{1}{n}$$

linearity of expectation

$E[|N|] = n \cdot p = \frac{1}{\varepsilon} \ln(mn)$ $\leftarrow$ "small size".

$$\frac{1}{\epsilon n} \ln mn \geq 1$$

$$\epsilon \leq \frac{\ln mn}{n}$$

if $\epsilon \leq \frac{\ln mn}{n} \rightarrow |N| = n$

$\epsilon > \frac{1}{\text{—}}$ $\quad |N| = \frac{1}{\epsilon} \cdot \ln(mn)$

VC-dim

$\epsilon = \frac{1}{2}$ $\quad 2 \ln(mn)$

$m = O(n^2)$

$|X| = n$ $\rightarrow$ $O(\ln n)$ $\quad O(1)$

If we want success prob $\geq 1 - \gamma$

$\gamma = 1/n$

then $p = \frac{1}{\varepsilon n} \ln \frac{n}{\gamma}$.

$Y = |N|$

$Y_i = 1$ if x_i is added to N.

$$Y = \sum_{i=1}^{n} Y_i$$

$$E[Y] = E\left[\sum_{i=1}^{n} Y_i\right] = \sum_{i=1}^{n} E[Y_i]$$

$$= \sum_{i=1}^{n} p = np$$

Markov's inequality: $\Pr[X \geq t \cdot E[X]] \leq \frac{1}{t}$.

$\Rightarrow$ Markov's inequality is very weak.

A much stronger inequality is "Chernoff's bound".

Theorem: [a key stmt in proba.]

$X_1 \dots X_n$ ← independent random variables.

$X_i \in \{0, 1\}$.

$\Pr[X_i = 1] = p_i$

$X = \sum_{i=1}^{n} X_i \Rightarrow E[X] = \sum_{i=1}^{n} p_i$.

$\delta \in (0,1)$: $\Pr[X \geq (1+\delta) E[X]] \leq e^{-\frac{\delta^2 E[X]}{3}}$.

$\delta \in (0,1)$: $\Pr[X \le (1-\delta)E[X]] \le e^{\frac{-\delta^2 E[X]}{2}}$

Application: Discrepancy. (X, R) — X: n elements; R: m sets

Goal: color each element of X with $+1$ or -1.

$$\chi : X \to \{-1, +1\}$$

s.t. for each $S \in R$, the # of -1 elements are close to the # of $+1$ elements.

$$\text{disc}_\chi(S) = \left| \sum_{x \in S} \chi(x) \right|$$

Theorem: Compute a coloring χ s.t.

$$\max_{S \in R} \text{disc}_\chi(S) = O(\sqrt{n \ln m})$$

Theorem (in words): there is a coloring χ s.t. for all sets $S \subseteq R$,

$$\left| \#\text{ of color in } S - \#\text{ of color in } S \right| \leq \sqrt{n \ln m}$$

Algorithm For $t = 1 \dots n$:

- color x_i with $+1$ or -1 with prob. $\frac{1}{2}$.

Theorem: With prob. $\geq \frac{9}{10}$, each set has discrepancy at most $\sqrt{n \ln m}$.

The probabilistic method.

P. Erdős.

Analysis:

(1) For each fixed $S \in R$, we will upper bound the probability that S has too large discrepancy.

(2) Use union bound to compute prob. that <u>some</u> set has too high a discrepancy.

Step 1: $S \in R$ $\quad Y_i = \begin{cases} 1 & \text{if } x_i \text{ colored } +1 \\ 0 & \text{if } x_i \text{ colored } -1 \end{cases}$

$$E\left[\overbrace{\sum_{x_i \in S} Y_i}^{Y}\right] = \sum_{x_i \in S} \underbrace{E[Y_i]}_{1/2} = \frac{|S|}{2}$$

We want:

$$Pr\left[\sum_{x_i \in S} Y_i \geq \frac{|S|}{2} + \Delta\right] \leq ?$$

$$\delta$$

$$-\left(\frac{2\Delta}{|S|}\right)^2 \cdot \frac{|S|/2}{3}$$

$$Pr\left[\sum_{x_i \in S} Y_i \geq \underbrace{\frac{|S|}{2}}_{E[Y]}\left[1 + \underbrace{\frac{2\Delta}{|S|}}_{\delta}\right]\right] \leq e$$

$$= e^{-\frac{\Delta^2}{|S|}\frac{2}{3}} \leq e^{-\frac{\Delta^2}{n}\frac{2}{3}}$$

(2)

$$\Pr\left[\text{Some set of } R \text{ has discrepancy} \geq \frac{|S_i|}{2} + \Delta\right] \leq m \cdot e^{-\frac{\Delta^2}{n}\cdot\frac{2}{3}}$$

We want to choose as small Δ as possible, so that the above prob. is $\leq \frac{1}{10}$

$$\Delta = \sqrt{6n \ln m}$$

$$m \cdot e^{-\frac{3\ln n \ln m}{n}\cdot\frac{2}{3}} = m e^{-\ln 10m} = \frac{1}{10}$$

$$\Pr\left[\text{some set has discrepancy} \geq \sqrt{\tfrac{3}{2} n \ln 10m}\right] \leq \frac{1}{10} \quad \cdots \quad [2]$$

$$\text{discrepancy} = O\left(\sqrt{n \ln m}\right)$$

$$\Pr\left[\text{we get discrepancy } O(\Delta) \text{ for all sets}\right] \leq m \cdot e^{\frac{-\Delta^2}{n}} < 1$$

Problem: (X, R) $\varepsilon \in (0,1]$ $A \subseteq X$ of size t randomly

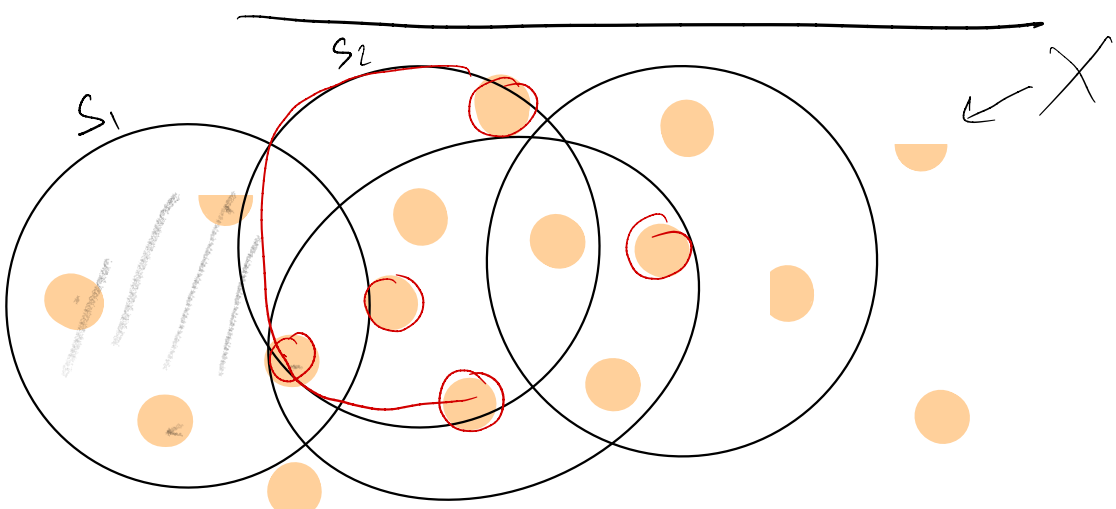
X: n elements; R: m sets.

↳ pick each $x_i \in X$ into A independently with prob. $\frac{t}{n}$.

$S \in R$: $E[|A \cap S|] = |S| \frac{t}{n}$.

$A \subseteq X$ is an ε-approximation if $\forall S \in R$, we have $|A \cap S| = \frac{|S| t}{n} \pm \underbrace{\varepsilon t}_{\text{error}}$.

Prove that $\exists$ an ε-approximation of size $O\left(\frac{1}{\varepsilon^2} \ln m\right)$.



A: picked each $x_i \in X$ w.p. t/n.

① Fix $S \in R$.

Y_i: 1 if x_i is picked, 0 otherwise.

$$|A \cap S| = \sum_{x_i \in S} Y_i$$

$$E[|A \cap S|] = \sum_{x_i \in S} E(Y_i) = |S| \cdot \frac{t}{n}$$

$$Pr\left[|A \cap S| \geq \frac{|S|t}{n} + \varepsilon t \right]$$

$$= Pr\left[|A \cap S| \geq \underbrace{\frac{|S|t}{n}}_{E} \left(1 + \underbrace{\frac{n\varepsilon}{|S|}}_{\delta} \right) \right]$$

$$\leq e^{-\left(\frac{\varepsilon n}{|S|}\right)^2 \cdot \frac{|S|t}{n} \cdot \frac{1}{3}}$$

$$= e^{-\frac{\varepsilon^2 n}{|S|} t + \frac{1}{3}}$$

$$= e^{-\frac{\varepsilon^2 t}{3}}.$$

$$\leq e^{-\frac{\varepsilon^2 t}{3}}$$

(2) $\Pr[\text{A fails for } \underline{\text{some}} \text{ set } R] \leq m \cdot e^{-\frac{\varepsilon^2 t}{3}}$

$$t = \frac{10}{\varepsilon^2} \ln m.$$

□